Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2019-20

Course Code	**Course Title**	**July-December** 2022	**Credit:** 03
CCE-323	Optical Fiber Communication		**Time:** 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A] What distinguishes an optical fiber communication system from a normal communication system? Using the relevant block diagram, describe the optical fiber communication system. 7

[B] An 8 km length of fiber with a mean optical power launch of 120 μW will have a mean optical power of 3 μW at the fiber output. 7

Assess:

i. If no connectors or splices are present, the total signal attenuation or loss in decibels via the fiber;

ii. The fiber's signal attenuation per kilometer.

iii. the total signal attenuation for a 20 km optical link made of the same fiber with 2 k m-separated splices that each provide a 2 dB attenuation;

iv. The numerical input/output power ratio in (c).

2 [A] What are the origins of fiber optic losses in silica? Give a thorough explanation of the intrinsic absorption loss mechanisms along with any required examples. 7

[B] The parameters of two step index fibers are as follows: 7

i. a multimode fiber with an operational wavelength of 0.82 μm, a relative refractive index difference of 3%, and a core refractive index of 1.500;

ii. a single-mode fiber with an 8 μm core diameter, an operating wavelength of 1.55 μm, a relative refractive index difference of 0.3%, and the same core refractive index as (a).

Estimate the critical radius of curvature at which large bending losses occur in both cases.

3 [A] i. What is optical fiber? Show its typical configuration with necessary diagram. 7

ii. Describe different advantages of optical fiber communication.

[B] Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7\times10^{-11} m^2N^{-1}$. The refractive index and the photo elastic coefficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm. Boltzmann's constant is $1.381\times10^{-23} JK^{-1}$. 7

$\lambda_R = \frac{8\pi^3 n^8 p^2 \beta_c K T_F}{3\lambda^4}$

4 [A.] What is meant by the fusion splicing of optical fibers? Explain the basic arc fusion method with pre-fusion method for accurately splicing optical fibers. Draw Electric arc fusion splicing diagram. 6

[B.] Briefly describe the major reasons for the cabling of optical fibers which are to be placed in a field environment. State the functions of the optical fiber cable. 3

[C.] Draw and explain a degenerate p–n junction where the position of the Fermi level and the electron occupation (shading) with applied bias and with no applied bias will be clearly depicted. 5

5 [A.] Define the two processes by which light can be emitted from an atom. Discuss the requirement for population inversion in order that stimulated emission may dominate over spontaneous emission. Illustrate your answer with an energy level diagram of a common non-semiconductor laser. 4

[B.] Explain mechanical splicing techniques that utilize grooves to secure optical fibers during the jointing process, and how these grooves contribute to the overall stability and alignment of the splice? 5

[C.] What is the main difference between LED and Laser? Draw and distinguish among different type of misalignment in fiber joint which are source of loss at a fiber-fiber connection. 5

6 [A.] Describe, with the aid of suitable diagrams, the major strategies and structures utilized in the fabrication of single-frequency injection lasers (The double-heterojunction injection laser). Indicate the reasons for the great interest in such devices. 5

[B.] A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device. 3

[C.] i. Draw schematic illustration of the vapor-phase deposition techniques used in the preparation of low-loss optical fibers. 6
ii. Briefly explain outside vapor-phase oxidation process.

8.88×10^{-6}
2.75×10^{-2}

Optical

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Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2018-2019

Course Code	Course Title	
CCE-323	Optical Fiber Communication	**Credit:** 03 **Time:** 03 Hr **Marks:** 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] Define following terms with respect to optical laws – 4
a) Critical angle and acceptance angle
b) Refractive index

[B.] Discuss the requirement for population inversion in order that stimulated emission may dominate over spontaneous emission. Illustrate your answer with an energy level diagram of a common non-semiconductor laser. 4

[C.] "Longitudinal misalignment gives significantly minor losses per unit displacement than the Lateral misalignment"- True or false, justify your statement. 2

[D.] Draw and explain p–n junction with strong forward bias hence stimulated emission is obtained in this region. 4

2 [A.] Explain the different types of rays in fiber optic. 4

[B.] Briefly discuss, with the aid of suitable diagrams, the following concepts in optical fiber transmission: 6
a) Modes in a planar guide
b) Goos–Haenchen shift;

0.99

[C.] A step index fiber has a core refractive index of 1.5 and a core diameter of 50 μm. The fiber is jointed with a lateral misalignment between the core axes of 5 μm. Estimate the insertion loss at the joint due to the lateral misalignment assuming a uniform distribution of power between all guided modes when: 4
a) there is a small air gap at the joint;
b) the joint is considered index matched.

168.52 $\frac{3\sqrt{11}}{50}$

3 [A.] Describe the three types of fiber misalignment which may contribute to insertion loss at an optical fiber joint with appropriate figure. 6

[B.] Derive the expression for electromagnetic mode theory for optical propagation. 4
i. Electromagnetic waves
ii. Phase and group velocity

[C.] Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device. 4

4 [A.] Write short note on ST connector, SC connector and FC connector. 5
[B.] Describe the operational principle of optical coupler using 2 X 2 Coupler. 3
[C.] Write the working principle of wavelength converter. 6

5 [A.] Describe Optical isolators and Optical circulators. 6
[B.] What is an Optical Switch? Describe the working procedure of optical switch. 6
[C.] How fiber communication work in terrestrial and subsea system? 2

6 [A.] Describe different types of optical filters. 6
[B.] What is an optical attenuator used for? What are the different types of optical attenuators? 5
[C.] What is optical multiplexing and demultiplexing? 3

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Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2017-2018

Course Code	**Course Title**	**July-December 2020**	**Credit: 03**
CCE-323	Optical Fiber Communication		**Time: 03 Hr**
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.67 and a cladding refractive index of 1.60. Determine: 7

- a) the critical angle at the core–cladding interface;
- b) the NA for the fiber;
- c) the acceptance angle in air for the fiber.

[B.] When the mean optical power launched into a 10 km length of fiber is 160 μW, the mean optical power at the fiber output is 4 μW. 7

Determine:

- a) the overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices;
- b) the signal attenuation per kilometer for the fiber.
- c) the overall signal attenuation for a 12 km optical link using the same fiber with splices at 1 km intervals, each giving an attenuation of 2 dB;
- d) the numerical input/output power ratio in (c).

2 [A.] i. Show typical fiber optic configuration with required schema. 7

ii. Describe the different benefits of fiber-based communication.

[B.] Two step index fibers exhibit the following parameters: 7

- a) a multimode fiber with a core refractive index of 1.500, a relative refractive index difference of 3% and an operating wavelength of 0.82 μm;
- b) an 8 μm core diameter single-mode fiber with a core refractive index the same as (a), a relative refractive index difference of 0.3% and an operating wavelength of 1.55 μm.

Estimate the critical radius of curvature at which large bending losses occur in both cases.

3 [A.] i. What are the differences between the overall communication system and the optical fiber communication system? 7

ii. Explain the fiber optic communication system with a suitable block diagram.

[B.] Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7\times10^{-11}m^2N^{-1}$. The refractive index and the photo-elastic coefficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm. Boltzmann's constant is $1.381\times10^{-23}JK^{-1}$. 7

4. [A.] Discuss optical fiber cable design with regard to:
a) fiber buffering;
b) cable strength and structural members

[B.] How does single-frequency device operation of the distributed feedback (DFB) laser happens in comparison with the Fabry–Pérot laser? 4

[C.] Describe two common methods used in the fabrication of three- and four-port fiber couplers. 4

5. [A.] Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device. 4

[B.] Describe what is meant by the fusion splicing of optical fibers. Discuss the advantages and drawbacks of this jointing technique. 4

[C.] A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device. 3

[D.] What are the direct and indirect Band Gap Semiconductors? 3

6. [A.] Draw and explain p–n junction with strong forward bias hence stimulated emission is obtained in this region. 4

[B.] Briefly describe in general terms liquid-phase techniques for the preparation of multicomponent glasses for optical fibers. 7

[C.] An optical fiber has a core refractive index of 1.5. Two lengths of the fiber with smooth and perpendicular (to the core axes) end faces are butted together. Assuming the fiber axes are perfectly aligned, calculate the optical loss in decibels at the joint (due to Fresnel reflection) when there is a small air gap between the fiber end faces. 3

Patuakhali Science and Technology University
Final Examination of B.Sc. Engg. (CSE), Level: 3 Semester: II
Course Code: CCE 323 Course Title: Optical Fiber Communication
Credit Hour: 3.00 Full Marks: 70 Duration: 03 Hours

[Figures in the right margin indicate full marks. **Split answering of any question is not recommended.** Write the full question number e.g. 5(B)(i) before the answer paragraph]

Answer any 5 of the following questions

16-17

1 A A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.57 and a cladding refractive index of 1.50. Determine: — Chap – 2 (2.1) 7

i. the critical angle at the core-cladding interface;
ii. the *NA* for the fiber;
iii. the acceptance angle in air for the fiber.

1 B What are the differences between the general communication system and optical fiber communication system? Explain optical fiber communication system with appropriate block diagram. – Chap – 1 7

2 A When the mean optical power launched into an 12 km length of fiber is 125 μW, the mean optical power at the fiber output is 5 μW. — Chap – 3 (3.1) 7
Determine:

i. the overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices;
ii. the signal attenuation per kilometer for the fiber.
iii. the overall signal attenuation for a 16 km optical link using the same fiber with splices at 1 km intervals, each giving an attenuation of 1 dB;
iv. the numerical input/output power ratio in (iii).

$\alpha_{dB} \cdot L$

$\frac{P_i}{P_o} = 10 \log\left(\frac{dB}{10}\right)$

2 B i. What is optical fiber? Show its typical configuration with necessary diagram. 7

ii. Explain in details the intrinsic absorption loss mechanisms with necessary illustrations. – Chap – 3

3 A "The intrinsic loss through a multimode fiber joint is independent of direction of propagation." State whether the given statement is true or false. Chap-5 (232 pg) 4
Draw and explain different causes of Radiative Losses when the joint fiber to fiber.

Chap-5

3 B A step index fiber has a core refractive index of 1.5 and a core diameter of 50 μm. The fiber is jointed with a lateral misalignment between the core axes of 5 μm. Estimate the insertion loss at the joint due to the lateral misalignment assuming a uniform distribution of power between all guided modes when: 6

Chap-5 Slide-15

(i) There is a small air gap at the joint; → (5.2) 224 pg
(ii) The joint is considered index matched. Slide – 19

3	C	How does single-frequency device operation of the distributed feedback (DFB) laser happen in comparison with the Fabry–Pérot laser? — Cha-C	4
4	A	Briefly describe the major reasons for the cabling of optical fibers which are to be placed in a field environment. State the functions of the optical fiber cable. — Chap-9 (21, 22 slide)	4
4	B	A ruby laser contains a crystal of length 4 cm with a refractive index of 1.78. The peak emission wavelength from the device is 0.55 μm. Determine the number of longitudinal modes and their frequency separation. — Chap-6 (6.2) 306 pg	3
4	C	Draw and explain p–n junction with no applied bias and with strong forward bias hence stimulated emission is obtained in this region. — Chap-6	7
5	A	What is the main difference between LED and Laser? Draw and distinguish among different type of misalignment in fiber joint which are source of loss at a fiber-fiber connection.	5
5	B	A graded index fiber with a parabolic refractive index profile ($\alpha = 2$) has a core diameter of 40 μm. Determine the difference in the estimated insertion losses at anindex-matched fiber joint with a lateral offset of 1 μm (no longitudinal or angularmisalignment). When performing the calculation assume: (i) The uniform illumination of only the guided modes (ii) The uniform illumination of both guided and leaky modes Chap-5 (5.3) 226 pg	5
5	C	"Photo detector is any device used to detect electromagnetic radiation and light-sensitive."- State whether the given statement is true or false. Write down the working principle of photo detector?	4
6	A	Define the two processes by which light can be emitted from an atom. Discuss the requirement for population inversion in order that stimulated emission may dominate over spontaneous emission. Illustrate your answer with an energy level diagram of a common non-semiconductor laser.	5
6	B	The radiative and non-radiative recombination lifetimes of the minority carriers in the active region of a double-hetero-junction LED are 60 ns and 100 ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87 μm at a drive current of 40mA.	4
6	C	Discuss optical fiber cable design with figure for multi-fiber cables with regard to three distinct types fiber buffering.	5

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) Level: 3 Semester: II Final Examination of Session 2015-16, July-Dec 2018

Course Code: CCE 323 Course Title: Optical Fiber Communication

Credit Hour: 3.00 Full Marks: 70 Duration: 03 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended. Write the full question number e.g. 4(A)(c) before the answer paragraph]

Answer any 5 of the following questions

1	A	Discuss about the structure and performance characteristics of multimode step index fibers.	5
1	B	Show the differences between multimode step index fiber and graded index fibers.	4
1	C	The Si-O bond has a theoretical cohesive strength of 3.4×10^6 psi which corresponds to a bond distance of 0.18nm. A silica optical fiber has an elliptical crack of depth 8nm at a point along its length. Estimate (a) the fracture stress in psi for the fiber if it dependents upon the crack. (b) the percentage strain at the break.	5
2	A	Show the typical structure of plastic clad fibers.	3
2	B	How misalignment occurs for jointing the optical fibers?	5
2	C	Discuss the intrinsic coupling losses at fiber joints. How misalignment loss is calculated for multimode fiber joints?	6
3	A	What are the requirements for an optical fiber emitter?	4
3	B	Show the absorption, spontaneous emission and stimulated emission with energy state diagram.	5
3	C	Discuss the energy band structure for optical emission of semiconductors.	5

$E = 7\times10^{10}$

Ans = 0.0557

4.63×10^{-3}

1.54491207

1.3176331

$\frac{3\times10^{-3}}{1\times10^{-3}}$

4 A A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.60 and a cladding refractive index of 1.57. Determine: 7

(a) the critical angle at the core–cladding interface;

(b) the NA for the fiber;

(c) the acceptance angle in air for the fiber.

4 B Two step index fibers exhibit the following parameters: 7

(a) a multimode fiber with a core refractive index of 1.45, a relative refractive index difference of 4% and an operating wavelength of 0.92 μm;

(b) a 10 μm core diameter single-mode fiber with a core refractive index the same as (a), a relative refractive index difference of 0.4% and an operating wavelength of 1.50 μm.

Estimate the critical radius of curvature at which large bending losses occur in both cases.

5 A Describe the electromagnetic spectrum showing the region used for optical fiber communications. 7

5 B Silica has an estimated fictive temperature of 1350 K with an isothermal compressibility of $8 \times 10^{-11} m^2N^{-1}$. The refractive index and the photoelastic coefficient for silica are 1.36 and 0.276 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.50, 0.75 and 1.00 μm. Boltzmann's constant is $1.381 \times 10^{-23} JK^{-1}$. 7

6 A What are the differences between the general communication system and optical fiber communication system? Explain optical fiber communication system with appropriate block diagram. 7

6 B When the mean optical power launched into an 10 km length of fiber is 140 μW, the mean optical power at the fiber output is 5 μW. 7

Determine:

a) the overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices;

b) the signal attenuation per kilometer for the fiber.

c) the overall signal attenuation for a 20 km optical link using the same fiber with splices at 2 km intervals, each giving an attenuation of 2 dB;

d) the numerical input/output power ratio in (c).

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) Level-3, Semester-II Final Examination-2016 (July-Dec)

Course Code: CIT 323 Course Title : Optical Fiber Communication

Credit Hour : 3.00 Full Marks:70 Duration: 3 Hours Session:2013-2014

Fiber

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

1. a) Explain the difference between the general communication system and optical fiber communication system using block diagram. 3
   b) What is optical fiber? Show its typical configuration with necessary diagram. 2
   c) Define the terms with suitable diagram: Critical angle, Phase velocity, Group velocity. 3
   d) Graphically show different ray propagation techniques in optical fiber. 2
   e) Develop two equations for numerical aperture that measure the light collecting ability of a fiber and show how the refractive index may be related to the numerical aperture. 4

2. a) Why the present form of optical communication is limited within the regions of 870nm, 1330nm and 1550nm? Explain with appropriate technical reasons. 4
   b) A step index fiber with a large core diameter compared with the wavelength of the transmitted light has an acceptance angle in air of 22° and a relative refractive index difference of 3%. Estimate the numerical aperture and the critical angle at the core-cladding interface for the fiber. 5 $n_1\sqrt{2\Delta}$
   c) What are the important factors to be considered for making a suitable joint between two fibers? 2
   d) What is group index? Draw basic structure of an optical receiver. 3

3. a) What are different dispersion mechanisms in fiber optic waveguide? Derive the mathematical expression for the material dispersion parameter. 4
   b) Suppose the digital bit pattern 1011 of the broadening of lightpulses as they are transmitted along a fiber as fiber input. Compare the fiber output at a distance for L_1 and L_2 ($L_1 < L_2$) 3
   c) What are the sources of losses in silica fiber optic? Explain in details the intrinsic absorption loss mechanisms with necessary illustrations. Also discuss fiber bending loss with proper diagram. 5
   d) Show different misalignment configurations that may occurs during connections between the fibers. 2

4. a) Describe the preparation method of optical fiber using liquid phase techniques. 4
   b) Write down the names of optical splices and connectors. Explain the applications of expanded beam connector. 4

Electron
Rayleigh

c) Two step index fibers exhibit the following parameters: 6

(i) A multimode fiber with a core refractive index of 1.5, a relative refractiveindex difference of 3% and an operating wavelength of 0.82μm;

(ii) An 8μm core diameter single-mode fiber with a core refractive index thesame as (i), a relative refractive index difference of 0.3% and an operatingwavelength of 1.55μm.

Estimate the critical radius of curvature at which large bending losses occur in both cases.

5. a) What are the different buffering techniques? Explain the Fresnel reflection mechanisms in an optical fiber joint. 3

b) What is hetero-junction? Explain how a concentrated beam of light is obtained by double hetero-structure of optical devices. 3

c) What are the advantages of LED? Show the basic structure of Laser and then explain the principles of light amplification in Laser cavity. 3

d) Explain why does the resposivity of a photo detector become shorter wavelengths and the beyond cutoff wavelength. 2

e) What do you mean by population inversion and pumping? 3

6. a) What is photo detector? Explain the photo detection principles in intrinsic material in brief. 3

b) Show that the responsivity is directly proportional to the quantum efficiency at a particular wavelength. 3

c) A p–nphotodiode has a quantum efficiency of 50% at a wavelength of 0.9μm. Calculate: 3

(a) its responsivity at 0.9μm;

(b) the received optical power if the mean photocurrent is 10^{-6}A;

(c) the corresponding number of received photons at this wavelength.

d) "The frequency response for an optical fiber system is more in optical than electrical bandwidth"- Justify the statement with relevant mathematical formula. 3

e) Show the performance comparison of ELED and SLED. Draw the V/I characteristics of PD including different regions of operation. 2

Patuakhali Science and Technology University

Final Examination of B.Sc. Engg (CSE) Level-3, Semester-2 July-December-2015

Course Code: CCE-323 Course Title: Optical Fiber Communication

Session :2012-2013 Credit Hour: 3.00 Full Marks: 70 Duration: 3.00 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 7 of the following questions

Fiber

1. a) Show the basic structure of an optical fiber. 2
   b) Illustrate different types of modulation technique. 3
   c) Compare between power efficiency and peak to average power ratio. 3
   d) Make differentiate between spectral efficiency and reliability. 2

2. a) Define photo detector. What is the difference between p-i-n photo detector and avalanched photodiode? 4
   b) Distinguish among shot noise, thermal noise and amplified spontaneous emission noise. 3
   c) Compare LED and semiconductor laser diode. 3

3. a) Define optical fiber sensor. What is the significance of optical fiber sensor? 3
   b) Show different types of fiber connector. 2
   c) Briefly describe different types of optical fiber sensor. 3
   d) Write short notes on fiber coupler, circulator and mode scrambler. 2

4. a) Show the drawing and preparation of optical Fiber. 3
   b) Illustrate different stages of optical fiber cabling with difference of dual core and six core fiber cables. 3
   c) Draw the outdoor and undersea cable design of optical fiber. 2
   d) Define fiber joint and joint loss. 2

5. a) Why optical amplification is needed? Write the general application of optical amplification. 3
   b) Write down the issues of amplifier design. Show the categories of EDFA. 4
   c) What is erbium doped fiber amplifier? Draw the basic EDFA amplifier design. 3

6. a) Differentiate between step index and graded index fiber. 3
   b) Show the comparison between mathematical expression of material dispersion and waveguide dispersion. 2
   c) Define the sources of noise of an optical receiver. 2
   d) Distinguish between optical direct detection receiver and coherent detection receiver. 3

7. a) Briefly define the Raman amplifier. 3
   b) Describe semiconductor optical amplifier (SOA) with its advantage and disadvantage. 3
   c) Narrate the semiconductor optical amplifier (SOA) operation. How WDM optical network works? 4

8. a) Write down the advantages and disadvantages of optical transmission. 2
   b) Show the fiber distributed data interface (FDDI) specification with ring structure. 3
   c) What do you mean by optical cross connect? Explain technologies for building a switch fabric of an O-C 3
   d) How optical Add-Drop Multiplexer work. 2